

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials.
- Show all of your work, and use correct notation. Failure to do these things will result in a deduction of points.
- You have **1 hour** to complete this exam.
- I'm proud of you—you've got this!

1. (8 pts.) With any hypothesis test, we could make the wrong decision. In the table below, determine the cells that correspond to a wrong decision, and write in these cells the type of error that is being made. Leave the other cells blank.

		Decision	
		Reject H_0	Fail to Reject H_0
Reality	H_0 is true		
	H_0 is false		

2. (12 pts.) Let $Y_1, \dots, Y_n$ be independent random variables, each with probability density function

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

Show that $\bar{Y}$ converges in probability to $2/3$.

3. (16 pts.) Let $Y_1, \dots, Y_n$ denote a random sample from a population whose probability density function is given by

$$f(y|\theta) = \begin{cases} \frac{3}{\theta} y^2 e^{-y^3/\theta}, & y > 0 \\ 0, & \text{otherwise} \end{cases}.$$

- (a.) (12 pts.) Use the Factorization Criterion to show that $\sum_{i=1}^n Y_i^3$ is a sufficient statistic for θ .

- (b.) (4 pts.) Is this sufficient statistic unique? Why or why not?

4. (28 pts.) Let $Y_1, \dots, Y_n$ constitute a random sample from the probability density function given by

$$f(y|\theta) = \begin{cases} \theta y^{\theta-1}, & 0 < y < 1, \quad \theta > 0 \\ 0, & \text{otherwise} \end{cases}.$$

- (a.) (12 pts.) Find the method of moments estimator, $\hat{\theta}_{MOM}$, for θ .

(b.) (12 pts.) Find the maximum likelihood estimator, $\hat{\theta}_{MLE}$, for θ .

(c.) (4 pts.) What is the MLE of $e^{-\theta}$? Justify your answer.

5. (24 pts.) Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$.

(a.) (12 pts.) Find a minimal sufficient statistic (MSS) for λ .

(b.) (12 pts.) Find a minimum variance unbiased estimator (MVUE) for λ . Justify all of your work.

Choose **one** of problems 6-8 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may randomly select which one to grade.

6. (12 pts.) Prove the following theorem:

Theorem 9.5: Rao-Blackwell Theorem. Let $\hat{\theta}$ be an unbiased estimator for θ such that $V(\hat{\theta}) < \infty$. If U is a sufficient statistic for θ , define $\hat{\theta}^* = E(\hat{\theta}|U)$. Then,

$$E(\hat{\theta}^*) = \theta \quad \text{and} \quad V(\hat{\theta}^*) \leq V(\hat{\theta}).$$

7. (12 pts.) Suppose $Y_1, \dots, Y_n$ are i.i.d. such that $E(Y_i) = \mu$, $V(Y_i) = \sigma^2$, $E(Y_i^2) = \mu'_2$, and $E(Y_i^4) = \mu'_4$ are all finite. Prove that $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2 \xrightarrow{p} \sigma^2$.

NOTE: You may use the fact that S_n^2 can be rewritten as

$$S_n^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}_n^2 \right].$$

8. (12 pts.) Prove the following theorem:

Theorem 9.1: If $\hat{\theta}_n$ is an unbiased estimator of θ and $\lim_{n \rightarrow \infty} V(\hat{\theta}_n) = 0$, then $\hat{\theta}_n$ is a consistent estimator of θ .